

BTC/USD Trading Strategy Backtesting Report

Strategy Logic :

I used ARIMA + GARCH (I didn't use SARIMA, because it was pretty clear that the series isn't seasonal) approach for the BTC/USD Trading Strategy, as ARIMA is the one of the most widely used approaches in time series forecasting, and before fitting series must be stationary or transformed using differencing. So first I used both ADF and KPSS test on the Box Cox transformed price series which confirmed that the price level is non-stationary and the first difference is stationary, (giving $I = 1$, for ARIMA)

Also ARIMA focuses on modelling the conditional mean of a time series and typically assume constant variance over time and financial returns do not satisfy this assumption as market volatility changes through time, that's why we use GARCH and most common model used is GARCH(1, 1).

For technical indicators I used SMA (50, 100, 200) and EMA (15) (inspired by the video provided in the notion notes) and MACD crossover state, RSI relative to the 50 midline, and Holt's linear trend model each cast a directional vote.

All indicators and model fits are walk-forward: the ARIMA, GARCH, Box-Cox lambda, and Holt models are re-estimated every 5 trading days on an expanding window containing only data up to and including the current day, with a 250-day burn-in period before any trade is taken. This structurally prevents lookahead bias, since no fit ever has access to data beyond the day being traded. Position exits are additionally protected by an ATR(14)-based trailing stop (2x ATR).

Key Performance Metrics :

Metrics	Value
Total Trades	60
Win Rate	45%
Long / Short Trades	31/29
Net Profit (on \$1,000)	\$3083.72
Benchmark Return (Buy & Hold)	\$236
Sharpe Ratio	0.84

Maximum Drawdown	60.65%
Average Win / Average Loss	\$398/\$232
Average Holding Time	14 days

Backtested on BTC/USD daily data (2018-2022, 1826 candles), starting capital \$1,000, compounding enabled, 0.15% transaction fee per trade.

ARIMA Order Selection :

ARIMA Order	Sharpe Ratio	Net Profit	Max Drawdown	Note
(0,1,0)	0.93	\$4,342	44.8%	Best
(1,1,1)	0.85	\$3,716	60.4%	-
(2,1,2)	0.63 - 0.80	\$1,562 - \$2,629	62 - 75%	Initial
(3,1,1)	N/A	N/A	N/A	Num (Kalm

An AIC/BIC comparison was done.

AIC alone was insufficient to select ARIMA order: it only measures in-sample fit quality on a fixed window and does not capture how estimation noise from extra parameters propagates through a walk-forward refitting process with volatility-scaled entry thresholds

I selected ARIMA(0, 1, 0) as it performed best when compared to others.

Limitations :

Daily BTC returns are mostly random walks, so the GARCH is the one which is providing the main returns, not ARIMA (as GARCH is a volatility filter unlike ARIMA)

Most of the profit came from the starting years (2019-2020) and my strategy actually lost money in 2021-2022, so the sharpe ratio may not hold for different periods